



# MARTÍN ARNALDO CERESA

## Formal Methods Engineer

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## TECH LANGUAGES

Haskell Lean4 Coq Agda

Go C Python Rust SML Erlang

Z CSP TLA/TLA+

## TECH STACK

ViM + Emacs XMonad Hakyll Git

LaTeX Markdown + Org GNU/Linux

## LANGUAGES

Spanish: Native | English: Fluent

Portuguese: Intermediate

## REFERENCES

### Prof. César Sánchez

IMDEA Software Institute  
cesar.sanchez@imdea.org

### Prof. Mauro Jaskelioff

FCEIA & IOG  
mjaskelioff@gmail.com

## PROJECTS

Termina 2023 - Ongoing

Transpiler from Termina to C plus a RTOS runtime for embedded systems. Designed and implemented in Haskell and runtime in C.

## EXPERIENCE

### Formal Methods Engineer | IO-G - ARC

05 2025 - Ongoing Madrid, Spain

- Mechanization of Midgard (L2 Optimistic Rollups) protocol in Lean4.
- L2 Tokenomics design.
- Data Availability Layer for L2 rollups design.
- Safety Analysis and Verification of Cardano smart-contracts.

### Postdoc Researcher | IMDEA Software Institute

03 2023 - 04 2025 Madrid, Spain

- Formalized Fraud Proof games in Lean.
- Designed offline monitoring properties for smart-contracts.
- Designed and implemented (in Haskell) Termina, a non-turing complete language for real-time critical embedded software.
- Developed a simulation environment for the Tezos blockchain smart contracts in Go.
- Guided PhD students and interns.

### Research Programmer | IMDEA Software Institute

10 2022 - 02 2023 Madrid, Spain

Implemented (in Rust) an intermediate layer expanding previous Tezos indexer features to monitor Tezos smart-contract with HLola.

### Visiting PhD Student | IMDEA Software Institute

08 2020 - 09 2022 Madrid, Spain

- Implemented a Coq library, Multi, to prove and reason about multi-contract interaction inspired on the computational model of the Tezos blockchain.
- Formalized (in Coq) different execution strategies, inspired by Tezos execution model, while introducing online and onchain smart-contract monitoring capabilities.
- Collaborated on the development of online monitors to blockchains.
- Collaborated in the implementation of Setchain (Byzantine tolerant grow-only sets).

### Undergraduate Intern | IMDEA Software Institute

03 2013 - 09 2013 Madrid, Spain

- Expanded the strategy toolbox of EasyCrypt, a modern proof assistant for cryptographers.
- Learned about modern cryptography proofs and game simulation-based proofs.
- Implemented (in OCaml) two different tactics: field/ring equality and automatic optimistic sampling.

FraudProofs 📅 2024 – Ongoing 🌐

Mechanization of Fraud-Proof arbitration games in L2 Optimistic Rollups. Implemented in Lean4, focused on proving honest players always win.

## Smart Multi Contract Interaction

📅 2021 – 2022 🌐

Designed and implemented a Coq library to reason about multi smart-contract interaction following the computational model of the Tezos Blockchain.

Haskell Lola 📅 2019 – 2020 🌐 🌐

Designed the core structure implementing HLola. Implemented the main abstraction lifting generic types of Haskell employed as data-theories in Lola.

QuickFuzz 📅 2016 – 2017 🌐

Out-of-the-box fuzzy generation of arbitrary files. Implemented Haskell arbitrary instances for arbitrary data structures in Haskell.

MegaDeTh 📅 2016 – 2017 🌐

Aggressive and generic instance derivation for recursive and mutually dependent Haskell data-types. Implemented entirely in Template Haskell.

EasyCrypt 📅 2013 🌐

Implemented tactics to simplify equations over rings and fields. Designed and implemented automatic optimistic sampling.

## EDUCATION

PhD in Informatics | UNR

📅 04 2015 – 04 2023 📍 Rosario, Argentina

- Thesis: Effectful Improvement Theory.
- Summary: An improvement theory for programming languages with algebraic effects.
- Advisor: PhD. Mauro Jaskelioff

Computer Science Degree | UNR

📅 03 2008 – 03 2015 📍 Rosario, Argentina

- Title: Simulation of Parallel Programs in Haskell
- Summary: A graphic library to observe and study the “parallel structure” of parallel programs in Haskell.
- GPA: 9.60

## TEACHING EXPERIENCE

Adjunct Professor | UNR

📅 2019 – 2021 📍 Rosario, Argentina

- Parallel and Concurrent Programs: I taught the basic problems and solutions of concurrent and parallel programming in C and Erlang.
- Compilers Course: I followed closely the book *Modern Compilers Implementation in ML* by Andrew Appel implementing a full-fledged compiler for the language *Tiger* without optimizations.

Teacher Assistant | UNR

📅 2011 – 2019 📍 Rosario, Argentina

My responsibilities included: assisting and conducting laboratory hours, crafting tests and final projects and grading them.

- Data Structures and Algorithms in C
- Compiler Theory and Practice
- Concurrent Programming in Erlang
- Introduction to pure programming languages and Haskell
- Parallel Thinking in Haskell
- Introduction to Lambda Calculus, STLC, Intermediate functional programming in Haskell
- Introduction to Intuitionistic Logic
- Introduction to Generalized Abstract Data Types

## COURSES

$\lambda$ -Calculus and Reasonable Cost Models 🌐 | UBA

📅 2019 📍 Buenos Aires, Argentina

DeepSpec Summer School 🌐 | Princeton University

📅 2018 📍 New Jersey, USA

Introduction to basic and algebraic Topology | UNR

📅 2018 📍 Rosario, Argentina

Haskell's Type System/GHC and its extensions 🌐 | ECI

📅 2018 📍 Río Cuarto, Argentina

FlexibleContexts, FlexibleInstances, GADTs, Phantom Types, RankN-Poly, DataKinds, MultiParamTypeClasses, FunctionalDependencies, TypeFamilies.

Introduction to Category Theory and Agda 🌐 | UNR

📅 2016 📍 Rosario, Argentina

Interactive Theorem Proving 🌐 | UBA

📅 2014 📍 Buenos Aires, Argentina

Introduction to Coq. Natural deduction, Curry-Howard correspondence and Lambda Calculus.

Introduction to Type Theory in Agda 🌐 | UNR

📅 2011 📍 Rosario, Argentina

# PUBLICATIONS

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## Journal Articles

- M. Capretto, **M. Ceresa**, A. F. Anta, A. Russo, and C. Sánchez, “Improving blockchain scalability with the setchain data-type,” *Distributed Ledger Technol. Res. Pract.*, vol. 3, no. 2, p. 12, 2024. DOI: 10.1145/3626963.
- M. Capretto, **M. Ceresa**, A. Fernández Anta, A. Russo, and C. Sánchez, “Improving blockchain scalability with the setchain data-type,” *Distrib. Ledger Technol.*, Oct. 2023, Just Accepted. DOI: 10.1145/3626963.
- **M. Ceresa** and M. Jaskelioff, “Effectful improvement theory,” *Sci. Comput. Program.*, vol. 217, p. 102 792, 2022. DOI: 10.1016/j.scico.2022.102792.
- G. Grieco, **M. Ceresa**, A. Mista, and P. Buiras, “Quickfuzz testing for fun and profit,” *Journal of Systems and Software*, vol. 134, pp. 340–354, 2017. DOI: 10.1016/j.jss.2017.09.018.

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## Conference Proceedings

- M. Capretto, **M. Ceresa**, A. F. Anta, P. Moreno-Sanchez, and C. Sánchez, “A secure sequencer and data availability committee for rollups,” in *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security, CCS 2025, Taipei, Taiwan, October 13-17, 2025*, C. Huang, J. Chen, S. Shieh, D. Lie, and V. Cortier, Eds., ACM, 2025, pp. 2129–2143. DOI: 10.1145/3719027.3765187.
- **M. Ceresa** and C. Sánchez, “Towards a mechanization of fraud proof games in lean,” in *6th International Workshop on Formal Methods for Blockchains, FMBC 2025, Hamilton, Canada, May 4, 2025*, D. Marmosier and M. Xu, Eds., ser. OA-Slcs, vol. 129, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025, 5:1–5:17. DOI: 10.4230/OASICS.FMBC.2025.5.
- J. Godoy, M. Capretto, **M. Ceresa**, J. P. Galeotti, D. Garbervetsky, C. Sánchez, and S. Uchitel, “Modal abstractions for smart contract validation,” in *28th ACM/IEEE International Conference on Model Driven Engineering Languages and Systems, MODELS 2025, Grand Rapids, MI, USA, October 5-10, 2025*, IEEE, 2025, pp. 175–185. DOI: 10.1109/MODELS67397.2025.00022.
- F. Gorostiaga, **M. Ceresa**, and C. Sánchez, “MOLA: A runtime verification engine factory by (meta-)interpreting embedded dsIs,” in *Practical Aspects of Declarative Languages - 27th International Symposium, PADL 2025, Denver, CO, USA, January 20-21, 2025, Proceedings*, E. Erdem and G. Vidal, Eds., ser. Lecture Notes in Computer Science, vol. 15537, Springer, 2025, pp. 53–70. DOI: 10.1007/978-3-031-84924-4\_4.
- A. Karmegam, G. L. Bianchi, M. Capretto, **M. Ceresa**, A. F. Anta, and C. Sánchez, “Setchain algorithms for blockchain scalability (extended abstract),” in *2025 IEEE International Parallel and Distributed Processing Symposium, IPDPS 2025 - Workshops, Milano, Italy, June 3-7, 2025*, IEEE, 2025, pp. 1298–1300. DOI: 10.1109/IPDPSW66978.2025.00219.
- M. Capretto, **M. Ceresa**, F. Gorostiaga, F. Macías, P. Pedregal, and C. Sánchez, “Offchain runtime verification (for the tezos blockchain),” in *Computer Security. ESORICS 2024 International Workshops - DPM, CBT, and CyberICPS, Bydgoszcz, Poland, September 16-20, 2024, Revised Selected Papers, Part I*, J. García-Alfaro et al., Eds., ser. Lecture Notes in Computer Science, vol. 15263, Springer, 2024, pp. 242–259. DOI: 10.1007/978-3-031-82349-7\_17.
- M. Capretto, **M. Ceresa**, and C. Sánchez, “Monitoring the future of smart contracts,” in *Fundamental Approaches to Software Engineering - 27th International Conference, FASE 2024, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2024, Luxembourg City, Luxembourg, April 6-11, 2024, Proceedings*, D. Beyer and A. Cavalcanti, Eds., ser. Lecture Notes in Computer Science, vol. 14573, Springer, 2024, pp. 122–142. DOI: 10.1007/978-3-031-57259-3\_6.
- M. Capretto, **M. Ceresa**, A. F. Anta, A. Russo, and C. Sánchez, “Setchain: Improving blockchain scalability with byzantine distributed sets and barriers,” in *IEEE International Conference on Blockchain, Blockchain 2022, Espoo, Finland, August 22-25, 2022*, IEEE, 2022, pp. 87–96. DOI: 10.1109/BLOCKCHAIN55522.2022.00022.
- M. Capretto, **M. Ceresa**, and C. Sanchez, “Transaction monitoring of smart contracts,” in *Runtime Verification*, T. Dang and V. Stolz, Eds., Cham: Springer International Publishing, 2022, pp. 162–180, ISBN: 978-3-031-17196-3.
- **M. Ceresa** and C. Sánchez, “Multi: A Formal Playground for Multi-Smart Contract Interaction,” in *4th International Workshop on Formal Methods for Blockchains (FMBC 2022)*, Z. Dargaye and C. Schneidewind, Eds., ser. Open Access Series in Informatics (OASlcs), vol. 105, Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022, 5:1–5:16, ISBN: 978-3-95977-250-1. DOI: 10.4230/OASlcs.FMBC.2022.5.
- **M. Ceresa**, F. Gorostiaga, and C. Sánchez, “Declarative stream runtime verification (hlola),” in *Programming Languages and Systems*, B. C. d. S. Oliveira, Ed., Cham: Springer International Publishing, 2020, pp. 25–43, ISBN: 978-3-030-64437-6.
- G. Grieco, **M. Ceresa**, and P. Buiras, “Quickfuzz: An automatic random fuzzer for common file formats,” in *Proceedings of the 9th International Symposium on Haskell, Haskell 2016, Nara, Japan, September 22-23, 2016*, G. Mainland, Ed., ACM, 2016, pp. 13–20. DOI: 10.1145/2976002.2976017.